1. Document argument analysis: finding and connecting arguments

The six-step ATIP classifier types one argument at a time. A real text contains many arguments that share statements and support each other. This note describes how to find those arguments in a document, record them as one graph, and hand each one to the classifier. The procedure is ours, not part of the published ATIP.

1.1. Background

The *Periodic Table of Arguments* (PTA) treats an argument as three statements: a **conclusion**, a **premise**, and a **lever** that says how the premise supports the conclusion (usually left unstated). The PTA's framework is *recursive*: any of the three can itself be argued for, and then becomes the conclusion of a further argument (Wagemans, PTA website; Section 1.10).

We record a text as a graph with two kinds of node: **statements** S and **connectors** K, one connector per argument. Three role relations link them:

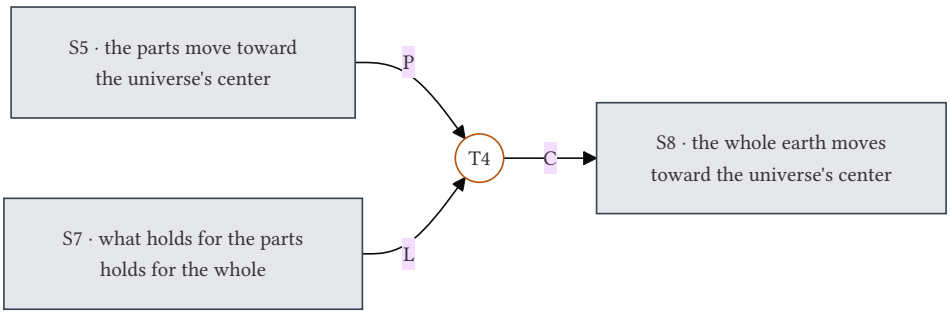
$$C, P, L \subseteq S \times K$$

$(s, k) \in C$ reads "s is the conclusion of k"; P is premise and L lever. Each connector has exactly one conclusion and one premise, and at most one lever statement:

$$|C^{-1}(k)| = |P^{-1}(k)| = 1, \quad |L^{-1}(k)| \leq 1$$

A statement has no role of its own. Its roles are the links it takes part in, so one statement can be the premise of one connector and the conclusion or lever of another. That is how the PTA's recursion shows up in the graph. A lever only gets a statement node when the text states it or argues about it; otherwise the classifier records it as a keyword.

In diagrams, statements are gray boxes and connectors are white circles; arrows run from premise and lever into the connector and from the connector to its conclusion:



The structure matches the Argument Interchange Format (AIF): statements are its information nodes and connectors its inference nodes, with the lever as an extra input (Chesñevar et al. 2006).

1.2. Running example

An Aristotelian argument about the earth's natural motion, as presented by Galileo (Finocchiaro's translation; see Section 1.10):

> Who is so blind as not to see that the parts of the element earth and of the element water, being heavy bodies, move naturally downwards, namely toward the center of the universe, assigned by nature herself as the goal and end of straight downward motion… Since this can be seen

> clearly, and since we are sure that the same holds for the whole as for the parts, must we not conclude that the natural motion of the [whole globe] earth is straight toward the center…?

We read it as Aristotle intends: "downwards" means toward the center of the *universe*, and the passage argues that the whole earth moves there. (Galileo's critique reads "the center" as the earth's own center, which makes the conclusion incoherent.) This main reading is called **R-expanded**: it has five arguments, T1–T5. Other readings are recorded as changes to it (Section 1.5.4).

1.3. Output

The procedure produces one YAML file per document with four parts:

- `statements`: each statement's canonical text, its source span, and whether it was `reconstructed` (implicit in the text).
- `candidates`: every place in the text that might signal an argument, with its **disposition**: what we decided it is.
- `connectors`: the arguments, as references to statements.
- `readings`: the main reading and each alternative, as a list of changes.

A disposition is one of:

disposition	meaning
`connector`	It signals an argument; names the connector.
`lever`	It introduces a stated lever of another connector.
`rephrase`	It restates something; no argument.
`conflict`	It attacks a statement ("but", "however"); say which in `note`. The attack's reason is an ordinary connector.
`background`	Explanation or context; no argument.
`rhetorical`	Framing only (a rhetorical question, emphasis); its content is kept, the framing dropped.
`unresolved`	Undecided; say why in `note`.

The example's file, with spans shortened:

```yaml
statements:
  S1:
    text: "The parts of earth and water are heavy bodies."
    span: "being heavy bodies"
  S2:
    text: "Heavy bodies move naturally downwards."
    reconstructed: true
  S3:
    text: "The parts of earth and water move naturally downwards."
    span: "the parts … move naturally downwards"
  S4:
    text: "Downward motion is motion toward the universe's center."
    span: "downwards, namely toward the center of the universe"
  S5:
    text: "The parts of earth and water move naturally toward the universe's
center."
    span: "the parts … toward the center of the universe"
  S6:
    text: "S5 can be seen clearly."
    span: "this can be seen clearly"
  S7:
    text: "What holds for the parts holds for the whole."
    span: "the same holds for the whole as for the parts"
  S8:
    text: "The whole earth moves naturally toward the universe's center."
    span: "the natural motion of the … earth is straight toward the center"
  S9:
    text: "Nature assigned the universe's center as the goal of downward motion."
    span: "assigned by nature herself as the goal …"

candidates:
  - {cue: "Who is so blind as not to see", disposition: rhetorical}
  - {cue: "being", disposition: connector, connector: T1}
  - {cue: "namely", disposition: connector, connector: T2}
  - {cue: "assigned by", disposition: connector, connector: T5}
  - {cue: "Since (1st)", disposition: connector, connector: T3}
  - {cue: "since (2nd)", disposition: lever, connector: T4, statement: S7}
  - {cue: "we are sure that", disposition: rhetorical}
  - {cue: "must we not conclude", disposition: connector, connector: T4}

connectors:
  T1: {conclusion: S3, premise: S2, lever_statement: S1}
  T2: {conclusion: S5, premise: S3, lever_statement: S4}
  T3: {conclusion: S5, premise: S6}
  T4: {conclusion: S8, premise: S5, lever_statement: S7}
  T5: {conclusion: S4, premise: S9}

readings:
  R-expanded: {}
  R-causal:
    note: "Heaviness itself is the reason; no general premise."
    replace: {T1: {conclusion: S3, premise: S1}}
    drop: [S2]
  R-rephrase:
    note: "'namely' only clarifies 'downwards'."
    merge: {S5: S3}
    drop: [T2, S4, T5, S9]
```

`lever_statement` is the optional lever node; the lever *keyword* is the classifier's output, stored separately. In INCEpTION, statements are `Statement` spans and connectors are `Connector` spans with `conclusion`, `premise` and `lever statement` links.

1.4. Principles

1. **Read the author's argument.** Use the author's meaning and terms, not a critic's, and don't strengthen the argument while reconstructing it.
2. **Find before you simplify.** List the claims and transitions before canonicalizing. Merging two descriptions too early can hide an argument.
3. **Look for the lever.** For each reason, ask what makes it relevant. A stated membership or likeness claim ("Socrates is a man", "robots are just like us") is a lever before it is a premise; the generalization it applies is the premise. See D3.
4. **Reuse statements.** One statement, one node, however many roles it plays.
5. **Add as little as possible.** Add an implicit statement only when an argument can't be stated without it, and mark it `reconstructed`.
6. **Keep readings apart.** Competing reconstructions are separate readings, not extra arguments in one graph.

1.5. Procedure

Five steps. Three are interpretive judgements; two are mechanical. A failed check in D5 sends you back to D2 for the affected part of the text.

step	kind	does	produces
D1	judgement	list the statements	`statements`
D2	judgement	find the cues that might signal arguments, and decide what each is	`candidates`
D3	mechanical	build a connector for each argument and link its statements	`connectors`
D4	judgement	record alternative reconstructions	`readings`
D5	mechanical	check each reading	a graph per reading

1.5.1. D1: statements

Produces: `statements`.

Go through the text and list every claim that could be a conclusion, premise or lever. Resolve pronouns and "this"/"so" where the reference is clear; record it in `note` where it isn't. Spans may overlap: a clause can contain a claim about another claim. Same wording doesn't make two statements the same, and different wording doesn't make them different.

Example. Nine statements (the YAML above). Three decisions matter:

- **S3 and S5 are separate**, though their spans overlap: "move downwards" and "move toward the universe's center" are the two descriptions "namely" links.
- **S2 is reconstructed**: it is the general premise "being heavy bodies" needs.
- **"we are sure that" is dropped** from S7: it is the arguer's stance, not part of the claim.

1.5.2. D2: candidates

Produces: `candidates`.

List every construction that might signal an argument, and give each a disposition. One sentence can hold several arguments, and one cue can signal none, so counting cues doesn't count arguments.

cue	may signal	check
because, since, therefore, so	a premise and a conclusion	Which claim is it for? Is it reasoning or just sequence?
being, given, relative clauses	a compressed premise or lever	Is it offered as a reason, or only describes?
namely, that is, by definition	a rephrase, or an identification the author is establishing	Would merging the two terms lose something?
generalizations, conditionals	a lever	Does it link two claims, or is it background?
clearly, can be seen, according to	a reason to accept a claim (δ)	Is it a reason, or emphasis?
rhetorical questions	an assertion or a challenge	What is actually being asserted?

For each candidate, ask: what is the reader asked to accept; what is offered as a reason; what makes the reason relevant; is any of these argued for elsewhere?

Is it a coherence marker? ATIP defines one by function only: it indicates which statement is the conclusion and which the premise, and it is not part of the propositional content (ATIP v6, Step 1). Step 2 replaces it with "because"; structure, illocutive and perlocutive markers are dropped. These three tests are ours; they reproduce ATIP's examples ("so", "if", and "many people have said" as a predicate, not a marker). An expression is a coherence marker if it passes all three:

1. **Substitution.** Replace it with "because" (transposing if needed). Both sides must still be complete statements, and nothing claimed may be lost. "Since", "so", "shows that", "proves that" pass; "many people have said q" fails, since "q because many people" has no premise left.
2. **Whose attitude.** If it reports an attitude to q, it is not a coherence marker. The arguer's own attitude is an illocutive or perlocutive marker ("unmistakably", "I argue that"): dropped, or A/U in δ. Anyone else's attitude ("according to The Economist", "we read there that") is content, part of a premise; it is δ only when the conclusion is the attributed proposition itself.
3. **What denying it means.** If denying it means denying that the premise supports the conclusion, it is a coherence marker: that is what "because" asserts. If it means denying a fact (did The Economist say so?), it is content.

A connector is anchored on its coherence marker. Where an illocutive marker sits next to it ("we might be led to conjecture that"), either anchor gives the same graph.

Example. Eight candidates (the YAML above). Five become connectors. The second "since" introduces S7, the stated lever of T4, not a sixth argument. The rhetorical questions are kept as the assertions S3/S5 and S8.

1.5.3. D3: connectors

Produces: `connectors`.

Create one connector per `connector` candidate and link its conclusion, premise and (if stated) lever statement. Then check every statement in the new connectors against the rest of the graph: is it argued for, or used, anywhere else? Add connectors only where the text supports them.

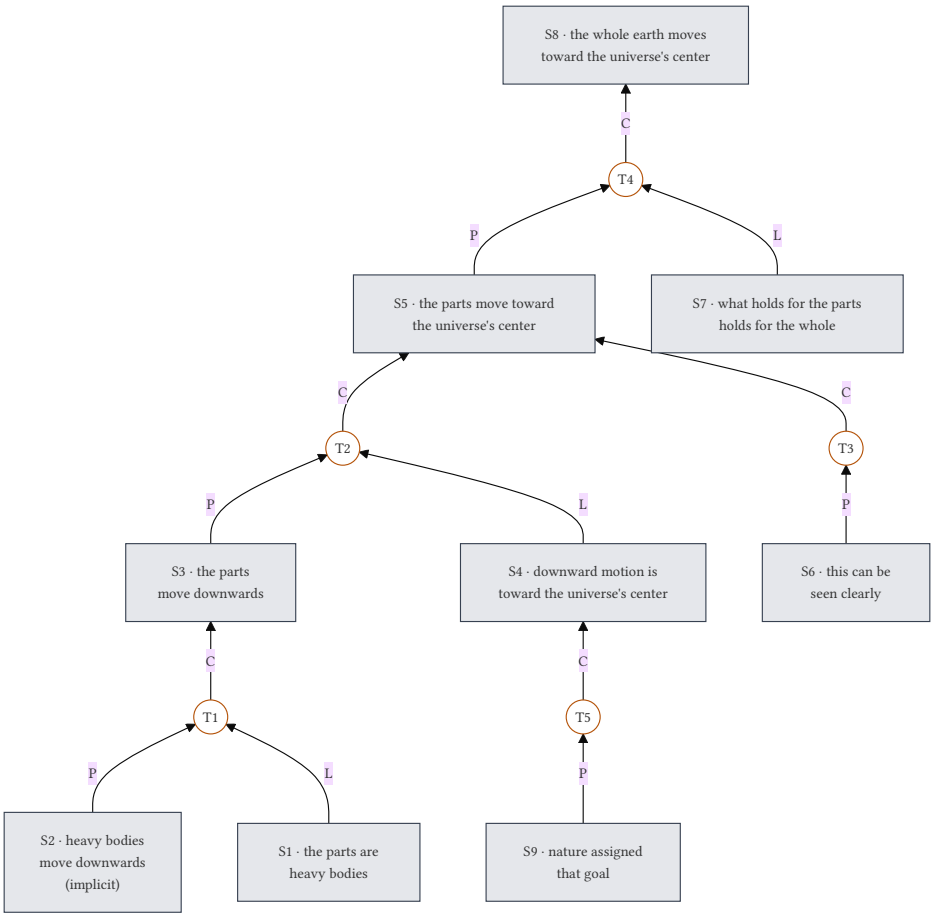
Premise or lever. When two statements jointly support a conclusion, decide which is the premise without looking at the type table:

1. If the text marks one as the reason ("because", "since", a participle such as "being heavy bodies"), it is the premise.

2. Otherwise, the generalization is the premise and the membership or likeness claim is the lever statement. “Socrates is mortal”: premise “all men are mortal”, lever “Socrates is a man” (β su G), not the reverse (α FF, no lever in the table).

The other split is mechanical (swap premise and lever), so the classifier reports its type as a runner-up; the labeller doesn’t weigh both. A connector typed `novel` is a sign the split is wrong: try the swap before keeping it.

Example. The result for R-expanded:



T1 ⟶ T2 ⟶ T4 is a chain: each conclusion is the next premise. T3 is a second argument for S5. T5 argues for S4, which is T2’s lever: the lever recursion the PTA describes.

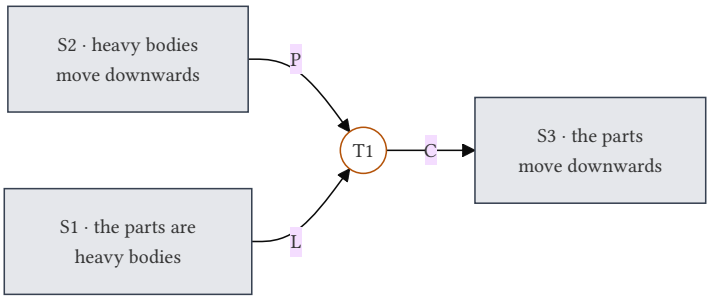
1.5.4. D4: readings

Produces: `readings`.

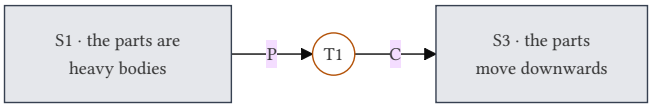
Where the text supports more than one reconstruction, keep the main reading and record each alternative as a change to it: connectors replaced or dropped, statements merged or dropped. Don’t add an alternative’s links to the main graph.

Example. “Being heavy bodies” has two readings:

R-expanded (genus): the general premise S2 supports S3; class membership S1 is the lever.



R-causal: heaviness S1 is itself the reason; S2 isn't needed.



Other open decisions:

decision	alternative	change
"namely"	It only clarifies "downwards".	R-rephrase: merge S5 into S3; drop T2, and with it S4, T5 and S9.
visibility	"This" refers to the downward motion only.	T3′s conclusion becomes S3.
nature	The assigned goal explains the motion instead of supporting S4.	T5′s conclusion changes, or T5 is dropped.

An alternative that only changes a connector's *type* (say, T5 as δ III Exp or α FF Cau) is a classification question, not a new reading.

1.5.5. D5: validate

Produces: a checked graph per reading.

Check, for each reading:

- every connector has one conclusion and one premise, and at most one lever statement;
- every reference points at an existing statement, and every `reconstructed` statement is used;
- every candidate has a disposition;
- no two connectors are the same argument;
- premises that only work jointly are noted: the schema has one premise per connector, so it can't show joint support. A connector stands for an entailment, and joint premises entail their conclusion together, not each on its own; see the argument graph in the ATIP design.

A failed check sends you back to D2 for the affected part of the text, not the whole document.

Example. R-expanded has 9 statements, 5 connectors and 13 links, all connected. In R-causal, nothing still uses S2; in R-rephrase, nothing still uses S4.

1.6. Traversal

Graph questions (what supports this claim, what uses it, which arguments are connected) follow links in both directions and in every role. With $E = C \cup P \cup L$, the neighbours of a statement and of a connector are

$$N(s) = \{k : (s,k) \in E\}, \quad N(k) = \{s : (s,k) \in E\}$$

and the connected parts of the document are found by a plain graph walk:

```
def components(graph):
    seen = set()
    for root in graph.nodes():
        if root in seen:
            continue
        pending, component = [root], []
        while pending:
            node = pending.pop()
            if node in seen:
                continue
            seen.add(node)
            component.append(node)
            pending.extend(graph.neighbours(node))
        yield component
```

This is connectivity, not the PTA's recursion: being connected doesn't mean one argument supports another. Walk each reading separately; mixing readings builds a graph nobody proposed.

Example. From any node, R-expanded's walk reaches all 14 nodes: one component.

1.7. Handoff to classification

Each connector, in each reading, is one input to the ATIP classifier: $a = (c, \rho, \mathrm{ctx})$ with c its conclusion, ρ its premise, and the document as context, plus the lever statement if there is one. Classify each connector once per reading; a statement shared by several connectors is not typed itself.

Example. T1 in R-expanded is (S3, S2) with lever S1; T1 in R-causal is (S3, S1), a different input with a different likely type (β pu G vs. α FF Cau).

Store each result against its reading and connector. A type's confidence is conditional on the reconstruction, so keep it separate from confidence in the reconstruction itself, and don't multiply type probabilities along the graph. If classification shows a reconstruction problem (a bad split, a missed middle step), go back to D4; don't change the graph just to get a listed type. `novel` is a valid result.

1.8. Stopping

Discovery stops when every candidate has a disposition and D5 passes. Missing support in the text is recorded as such ("no argument found"), not filled in. A reference to text outside the excerpt is `unresolved`.

1.9. Evaluation

Measure three things separately: finding arguments (candidates and their dispositions), reconstructing them (statement identity, roles, readings), and classifying them. A wrong role is a different error from a wrong type. Where the text supports several readings, gold labels should include them; a reading isn't better for having more arguments.

The judgements in D1, D2 and D4 are new semantic tasks, outside the classifier's oracles, and need their own baselines. Validation and traversal are mechanical.

1.10. References

- Wagemans, J.H.M. (2026). *Argument Type Identification Procedure (ATIP) – Version 6.* https://www.periodic-table-of-arguments.org/argument-type-identification-procedure

- Wagemans, J.H.M. *Periodic Table of Arguments*: theoretical framework. https://periodic-table-of-arguments.org/theoretical-framework/
- Chesñevar, C., McGinnis, J., Modgil, S., Rahwan, I., Reed, C., Simari, G., South, M., Vreeswijk, G., Willmott, S. (2006). "Towards an Argument Interchange Format." *Knowledge Engineering Review* 21(4), 293–316. https://doi.org/10.1017/S0269888906001044
- Galileo Galilei. *Galileo on the World Systems: A New Abridged Translation and Guide*, trans. and ed. M.A. Finocchiaro. University of California Press, 1997, pp. 83–84. Source of the running example.
- Hinton, M., Wagemans, J.H.M. (2023). "How persuasive is AI-generated argumentation?" *Argument & Computation* 14(1), 59–74. https://doi.org/10.3233/AAC-210026 Picks arguments out of a text by "a preliminary reading"; this procedure makes that step explicit.
- `pta-atip-algorithm.typ`: the six-step classifier this procedure feeds.